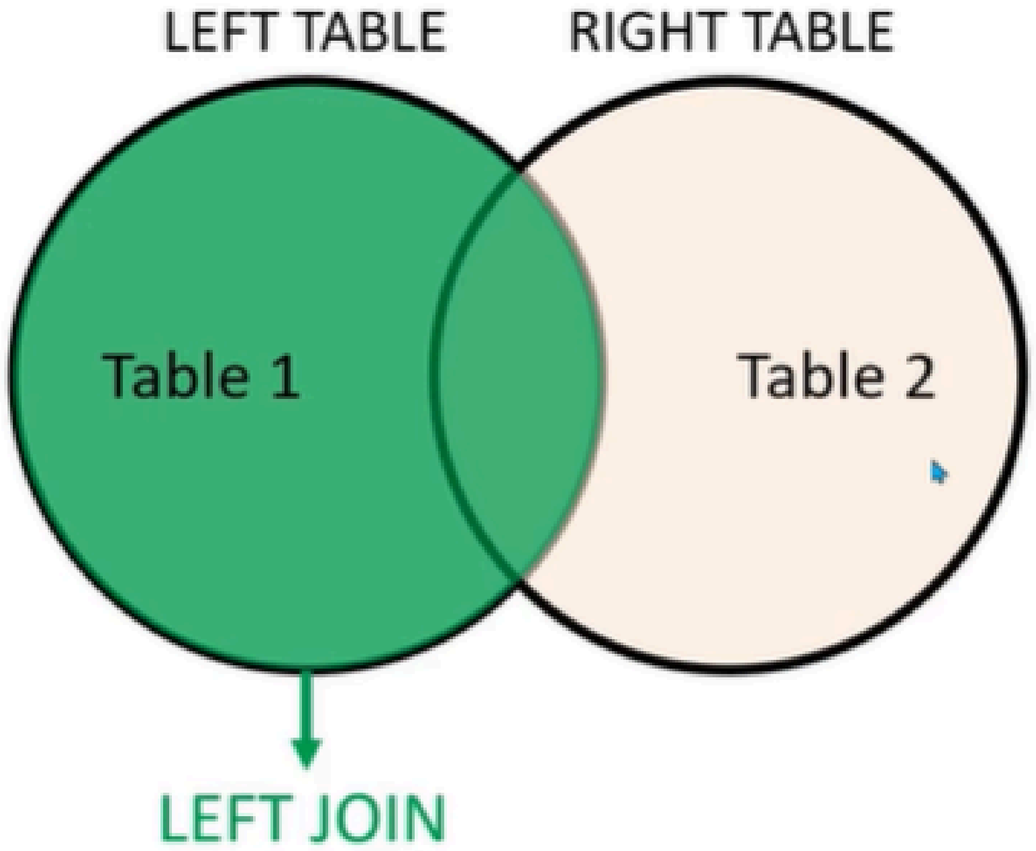


LEFT JOIN & RIGHT JOIN

TYPES OF JOINS IN MYSQL

- LEFT JOIN
- RIGHT JOIN
- CROSS JOIN

What is LEFT JOIN ?



The LEFT JOIN returns all records from the left table (table1),

and the matched records from the right table (table2)

LEFT JOIN Two Tables

Order Table

Order_Id	Product_Category_Id
101	2
102	3
103	1
104	2
105	**6**

Product Category Table

Category_Id	Category_Name
1	Electronics
2	Appliances
3	Mobile Devices
4	Apparel
5	Beauty





LEFT JOIN Result

Order_Id	Category_Name	Category_Id
101	Appliances	2
102	Mobile Devices	3
103	Electronics	1
104	Appliances	2
105	*NULL*	*NULL*

LEFT JOIN Syntax

```
SELECT columns

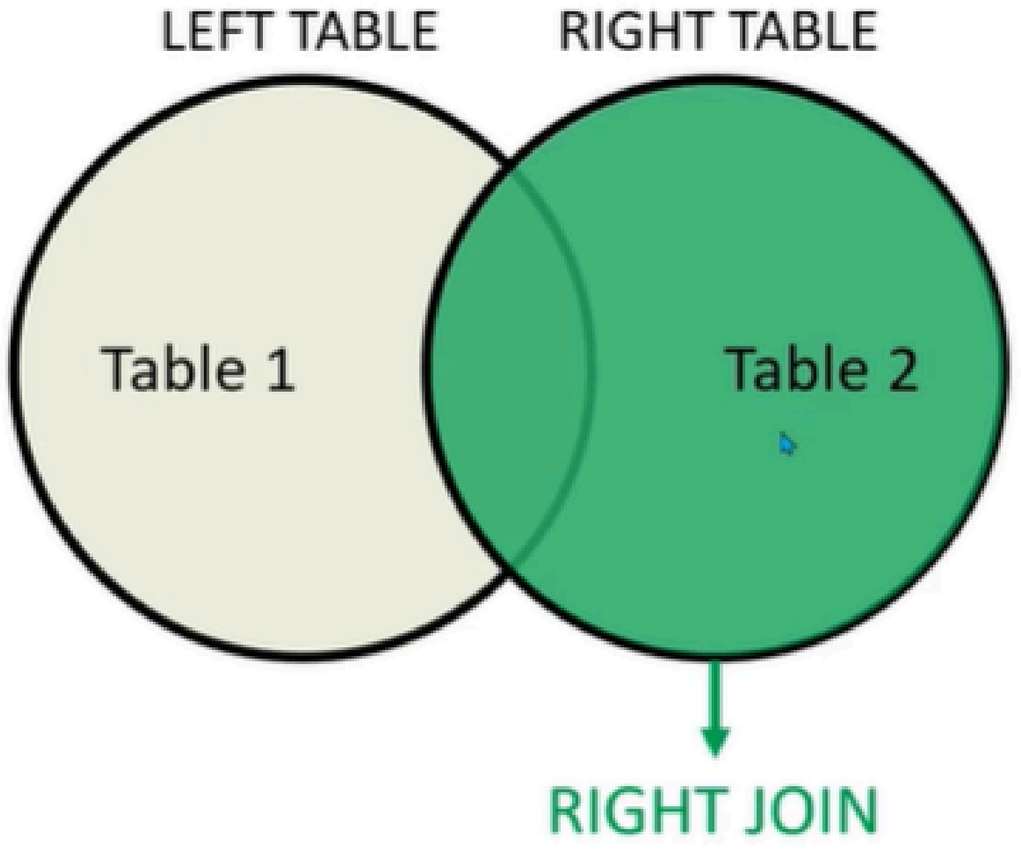
FROM table1

LEFT JOIN table2

ON table1.column_name = table2.column_name;
```



What is RIGHT JOIN ?



The RIGHT JOIN returns all records from the right table (table2),

and the matched records from the left table (table1).

RIGHT JOIN Syntax

```
SELECT columns

FROM table1

RIGHT JOIN table2

ON table1.column_name = table2.column_name;
```

FOREIGN KEY

PRIMARY KEY


MySQL™
CROSS JOIN

CROSS JOIN Two Tables

Order Table

Order_Id	Product_Category_Id
101	2
102	3
103	1
104	2
105	6

Cartesian Product

Product Category Table

Category_Id	Category_Name
1	Electronics
2	Appliances
3	Mobile Devices
4	Apparel
5	Beauty

CROSS JOIN Result

(Cartesian Product)

Order_Id	Category_Name	Category_Id		
101	Electronics	105	Electronics	1
102	Appliances	105	Appliances	2
101	Mobile Devices	105	Mobile Devices	3
101	Apparel	105	Apparel	4
...	...	105	Beauty	5

(Total Rows: 25)

Returns the Cartesian product of the two tables. No matching condition is required.

CROSS JOIN Syntax

```
SELECT columns

FROM table1

CROSS JOIN table2;
```


MySQL
GROUP BY &
HAVING Clause

GROUP BY Clause Demonstration

Order Items Table
(Aggregate Source)

Item_Id	Product_Type	Quantity
101	T-Shirt	10
102	Jeans	5
103	T-Shirt	2
104	Sneakers	2
105	Jeans	10
106	T-Shirt	8
107	Boots	4

Group & Aggregate

Sales Summary Table
(Grouped Result)

Product_Type	Total_Quantity (SUM)
T-Shirt	20 (sum of 10+2+8)
Jeans	15 (sum of 5+10)
Sneakers	2
Boots	4

GROUP BY and SUM Result
(Consolidated Summary: 4 Product Types)

Product_Type	Total_Quantity
T-Shirt	20 (sum of 10+2+8)
Jeans	15 (sum of 5+10)
Sneakers	2
Boots	4

Groups rows that have the same values and summarizes data into a consolidated view, like 'T-Shirt, 20'.

SELECT with GROUP BY Syntax

```
SELECT columns
FROM table_name
WHERE condition
GROUP BY column_name(s);
```

SELECT with GROUP BY Syntax

```
SELECT columns
FROM table_name
WHERE condition
GROUP BY column_name(s);
```

SELECT with GROUP BY Syntax

```
SELECT columns

FROM table1 INNER JOIN table2

ON table1.column_name = table2.column_name

WHERE condition

GROUP BY column_name(s);
```

HAVING Clause Demonstration: Filter on Grouped Data

Order Items Table
(Aggregate Source)

Item_Id	Product_Type	Quantity
101	T-Shirt	10
102	Jeans	5
103	T-Shirt	2
104	Sneakers	2
105	Jeans	10
106	T-Shirt	8
107	Boots	4



Group & Aggregate

GROUP BY and **SUM** (as in original)

Sales Summary Table
(Grouped Result)



Product_Type	Total_Quantity (SUM)
T-Shirt	20 (sum of 10+2+8)
Jeans	15 (sum of 5+10)
Boots	4

HAVING Clause Filtered Result:
(Only Groups with Total_Quantity > 5)

(Total Product Types: 3)





Product_Type	Total_Quantity
T-Shirt	20 (sum of 10+2+8)
Jeans	15 (sum of 5+10)
Boots	4

Groups rows with same values and summaries data. The HAVING clause *then filters* these final group summaries, removing any group that doesn't meet the condition (e.g., Sneakers were removed as Total_Quantity 2 < 5).

SELECT with GROUP BY & HAVING Syntax

```
SELECT columns
FROM table_name
GROUP BY column_name(s)
HAVING condition;
```


Thank
you